

Electronics

1) Electrons and charge

Electricity is movement of charge (electrons)

Atom $\rightarrow$ basic unit of matter

$\downarrow$
protons (+)

electrons (-)

neutrons (0) $\rightarrow$ no charge

Electrons move around the nucleus

Protons and neutrons stay in it.

+/+ or -/- $\rightarrow$ repel

+/- $\rightarrow$ attract

Electricity happens when electrons start flow

2) Electric current

Electric current is the flow of charge

Unit $\rightarrow$ Amperes (A)

Charges $\rightarrow$ (+) (-)

When conductor is exposed to a potential difference, charge carries the

3) Electrical Circuit

Electricity flows only in a complete circuit (closed loop)

Electrons must return to the starting point. If circuit is open $\rightarrow$ no flow
Components:

Energy source $\rightarrow$ provides power

Conductors (wires) $\rightarrow$ path for current

Load $\rightarrow$ uses electricity

Switch $\rightarrow$ controls the circuit

Current, Flow:

Current $\rightarrow$ flow of electrons

Electrons have negative charges

moves from negative to positive

Current

Opposite to electrons

moves from positive to negative

Current $\rightarrow$ Ammeter (series)

4) Electric Charge

Current can also be defined as the charge passing through a circuit per unit time

Charge $\rightarrow$ coulombs (C)

$$Q = It$$

$Q \rightarrow$ charge (C)

$I \rightarrow$ current (A)

$t \rightarrow$ time (s)

5) Potential Difference

Potential difference is the amount of energy transferred per unit charge as it moves from one point to another in circuit

P. Difference $\rightarrow$ volts (V) $V = \frac{1}{C}$

Voltmeter (Parallel)

$$V = \frac{W}{Q}$$

$W \rightarrow$ work done

$Q \rightarrow$ charge



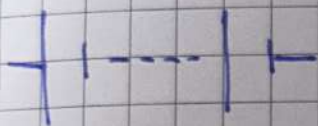
cell



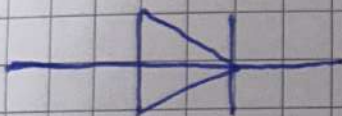
potentiometer



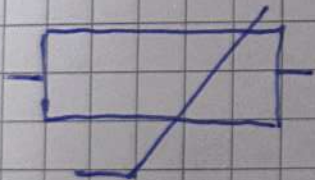
variable resistor



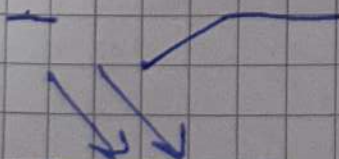
battery of cells



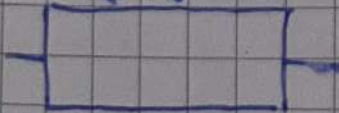
diode



thermistor



switch



light-dependent



lamp



capacitor

Electromotive force

When charge passes through a power supply it gains energy

e.m.f is amount of energy transferred per coulomb of charge when charge passes through a power supply.

measured $\rightarrow$ Volts (V)

$$\text{e.m.f} = \frac{\text{energy transferred}}{\text{charge}}$$

e.m.f = potential difference.

when cell got no current

high-resistance voltmeter around the terminals of the cell in an open circuit.

$$\mathcal{E} = IR + Ir = I(R + r)$$

$$\mathcal{E} = V - Ir$$

lost volts Ir

Series



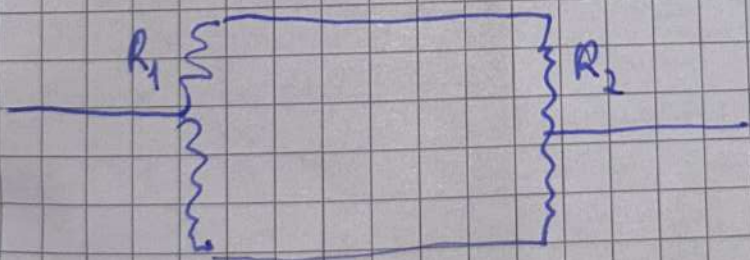
$$R_{\text{am}} = R_1 + R_2 + R_3$$

$j \rightarrow \text{const.}$

$U = R \cdot \overset{\text{düz}}{\text{asıldevre}}$

$$U_1 + U_2 + U_3 = U_{\text{am}}$$

Parallel



$V \rightarrow \text{const.}$

$$\frac{1}{R_{\text{um}}} = \frac{R_1 + R_2}{R_1 \cdot R_2}$$

Kirchoff's laws

Kirchoff's first law; The sum of the current entering a junction is always equal to the sum of the currents leaving the same junction

- A junction is a point of the currents where three circuit paths meet
- A Branch is a path connecting two junctions

$$I = I_1 + I_2 + I_3$$

Second law

The total emf in a closed circuit is equal to the sum of the potential differences across the components

$$E_1 + E_2 = V_1 + V_2 = V_3 + V_4$$